4️⃣

Phase 4 - São Paulo Finalization Code (Sanitized)

This phase accepts the TGW peering from Tokyo and adds routes for cross-region connectivity.

⚠️ **DEPENDENCY:** Phase 4 can only run AFTER Phase 3 completes. The peering attachment must be in `pendingAcceptance` state.

CIDR Reference

Region	Resource	CIDR
Tokyo	VPC	`10.0.0.0/16`
Tokyo	Private Subnets	`10.0.11.0/24`, `10.0.12.0/24`
São Paulo	VPC	`10.1.0.0/16`
São Paulo	Private Subnets	`10.1.11.0/24`, `10.1.12.0/24`

Step 1: Add Peering Accepter to `04-P-main-saopaulo-tgw.tf`

ADD to existing file:

```
##############################################
# ACCEPT TGW PEERING FROM TOKYO
# Liberdade accepts corridor from Shinjuku
##############################################
```

```
data "aws_ec2_transit_gateway_peering_attachment" "from_toky
o" {
  count = var.enable_tgw && var.tokyo_tgw_id != "" ? 1 : 0

  filter {
    name   = "transit-gateway-id"
    values = [aws_ec2_transit_gateway.liberdade_tgwlab3[0].i
d]
  }

  filter {
    name   = "state"
    values = ["pendingAcceptance", "available"]
  }
}

resource "aws_ec2_transit_gateway_peering_attachment_accepte
r" "accept_tokyo_peer01" {
  count = var.enable_tgw && var.tokyo_tgw_id != "" ? 1 : 0

  transit_gateway_attachment_id = data.aws_ec2_transit_gatewa
y_peering_attachment.from_tokyo[0].id

  tags = {
    Name    = "liberdade-accept-tokyo-peer01"
    Type    = "Cross-Region-Peering"
    Project = var.project_name
  }
}

##############################################
# TGW ROUTE TO TOKYO
# Route Tokyo CIDR (10.0.0.0/16) via peering
##############################################

resource "aws_ec2_transit_gateway_route" "liberdade_route_to_
```

```
tokyo" {
  count = var.enable_tgw && var.tokyo_tgw_id != "" ? 1 : 0

  destination_cidr_block         = var.tokyo_vpc_cidr  # 10.
0.0.0/16
  transit_gateway_attachment_id  = aws_ec2_transit_gateway_pe
ering_attachment_accepter.accept_tokyo_peer01[0].transit_gate
way_attachment_id
  transit_gateway_route_table_id = aws_ec2_transit_gateway.li
berdade_tgwlab3[0].association_default_route_table_id

  depends_on = [aws_ec2_transit_gateway_peering_attachment_ac
cepter.accept_tokyo_peer01]
}
```

Step 2: Create `04-P-main-saopaulo-routes.tf` (NEW FILE)

What it does: Adds VPC route so São Paulo private subnets can reach Tokyo via TGW.

```
##############################################
# SÃO PAULO VPC ROUTES TO TOKYO
# Liberdade private subnets → Tokyo for RDS access
# Destination: 10.0.0.0/16 (Tokyo VPC) → TGW
##############################################

resource "aws_route" "liberdade_to_tokyo_route01" {
  count = var.enable_tgw ? 1 : 0

  route_table_id         = aws_route_table.liberdade_private_
rtlab3.id
  destination_cidr_block = var.tokyo_vpc_cidr  # 10.0.0.0/16
  transit_gateway_id     = aws_ec2_transit_gateway.liberdade_
```

```
tgwlab3[0].id
}
```

Step 3: Update São Paulo Outputs

ADD to existing `05-outputs.tf` file:

```
output "saopaulo_peering_accepter_id" {
  description = "Sao Paulo peering accepter attachment ID"
  value       = var.enable_tgw && var.tokyo_tgw_id != "" ? aw
s_ec2_transit_gateway_peering_attachment_accepter.accept_toky
o_peer01[0].id : null
}

output "saopaulo_tgw_route_to_tokyo" {
  description = "TGW route to Tokyo CIDR"
  value       = var.enable_tgw && var.tokyo_tgw_id != "" ? aw
s_ec2_transit_gateway_route.liberdade_route_to_tokyo[0].desti
nation_cidr_block : null
}
```

Deployment Commands (Phase 4)

```
cd lab3/saopaulo

# Plan
terraform plan -out=phase4.tfplan

# Should show:
# - 1 data source (peering attachment lookup)
# - 1 peering accepter
# - 1 TGW route to Tokyo (10.0.0.0/16)
# - 1 VPC route to Tokyo (10.0.0.0/16)
```

```
# Apply
terraform apply phase4.tfplan
```

Verification Checklist (Phase 4)

- [ ] Peering accepted (state: `available` in both regions)
- [ ] São Paulo TGW route table has entry for Tokyo CIDR (`10.0.0.0/16`)
- [ ] São Paulo VPC route table (`liberdade_private_rtlab3`) has route to `10.0.0.0/16`

```
# Verify peering is available
aws ec2 describe-transit-gateway-peering-attachments --region
sa-east-1 \
  --query "TransitGatewayPeeringAttachments[?State=='availabl
e']"

# Verify TGW route to Tokyo
aws ec2 describe-transit-gateway-route-tables --region sa-eas
t-1

# Verify VPC route to Tokyo (10.0.0.0/16)
aws ec2 describe-route-tables --region sa-east-1 \
  --filters "Name=tag:Name,Values=*liberdade-private*" \
  --query "RouteTables[*].Routes[?DestinationCidrBlock=='10.
0.0.0/16']"
```

Post-Phase 4: Uncomment Tokyo TGW Route

🔗 **IMPORTANT:** After Phase 4 is applied and peering shows `available`, go back to Tokyo and uncomment the TGW route.

File: `04-3-tokyo.tf`

UNCOMMENT this block:

```
resource "aws_ec2_transit_gateway_route" "shinjuku_route_to_l
iberdade" {
  count                          = var.enable_tgw && var.saop
aulo_tgw_id != "" ? 1 : 0
  destination_cidr_block         = var.saopaulo_vpc_cidr  # 1
0.1.0.0/16
  transit_gateway_attachment_id  = aws_ec2_transit_gateway_pe
ering_attachment.shinjuku_to_liberdade_peer01[0].id
  transit_gateway_route_table_id = aws_ec2_transit_gateway.sh
injuku_tgwlab3[0].association_default_route_table_id

  depends_on = [aws_ec2_transit_gateway_peering_attachment.sh
injuku_to_liberdade_peer01]
}
```

Then run:

```
cd lab3/tokyo
terraform plan -out=phase4b.tfplan
terraform apply phase4b.tfplan
```

Traffic Flow Summary

```
São Paulo EC2 (10.1.11.x)
         |
         ▼
liberdade_private_rtlab3 → route 10.0.0.0/16 → TGW
         |
         ▼
liberdade_tgwlab3 (São Paulo) ——— TGW Peering ——— shinjuku_tg
wlab3 (Tokyo)
                                                        |
```

```
                                                    ▼
                                                helga_private_r
tlab3
                                                    |
                                                    ▼
                                                helga_rdslab3
(10.0.x.x)
```

✅ **Phase 4 Complete:** Cross-region connectivity is now established. São Paulo EC2 instances in private subnets can reach `helga_rdslab3` in Tokyo, but **NO PHI is stored outside Japan**.